The Steerwish: Intent Kept on Live Systems

Matthias Tafelmeier
Independent Scientist
Hof, Germany
ORCID: 0009-0005-7399-5584

Abstract

Autonomous agents that tend a shared substrate can measure it while it runs: couplings are discovered, response curves carry error bars, and the curves compose across hops. This paper asks what such a map is worth when the world is not still. The answer is the *steerwish* — intent made keepable: a promise to a live system, declared by its holder, kept against drift, and refused — not faked — when the measured range says no. Four claims, each carried by receipts from a live measurement campaign. *The map can steer*: through a four-line promise grammar, composed predictions land readings inside promised bands, repeating across seed runs — and the boundary is stated as measured, a limit of map density rather than of principle. *Refusal*: a target outside the measured range is declined by name, with zero actuation. *Drift survival*: a declared drift event is survived end to end — misses stamp, the map re-opens, the reading re-lands — with stable numbers across runs. *Upkeep economics*: a three-flight envelope prices the keeping in rounds and waterings, read absolutely between what luck alone keeps and what perfect hindsight could have kept, with the envelope's two failures reported as the honest edges of the economics. Keeping is priced in rounds and waterings, never against external keepers: the comparison is a named follow-up.

Keywords: autonomous agents, multi-agent systems, steerwish, promises, refusal, drift, maintenance cost, model-based control, system identification, self-adaptive systems, live systems

1 Introduction

When autonomous agents tend the same substrate, each one's parameter changes travel through the substrate into the others' readings. Prior work established that this hidden coupling is measurable without central observers and without stopping the system: edges are discovered by guardrail-bounded probing between the agents themselves [1], each edge yields a response curve with per-point error bars [2], and measured curves compose across nonlinear hops and converging paths [3]. The map that results is a certified, priced model of a live world.

This paper asks the question that line of work ends at: can the measured map — the substrate's *connectome*: every coupling edge with its measured response curve — *do* anything? Specifically — can a promise be held about the far end of the substrate, through the map, while the substrate moves? Holding a promise is not an optimum-seeking problem. It is a keeping problem: the reading must stay inside a band, the world drifts, and the keeping is work. Work has a price. And some promises, measured honestly, cannot be kept at all — the keeper that cannot see far enough must say so and touch nothing.

The claims, in the order the campaign receipts them:

1. **The map can steer.** Invert once through the composed map, act once, land inside the promised band; repeats across seeds show it is not luck. The boundary is stated as measured, not assumed away (§5.1).

2. **Refusal.** A target outside the measured range is refused by name, with zero actuation — knowing what you do not know, under action (§5.2).
3. **Drift survival.** The promise survives a declared drift event: misses stamp, the map re-opens, the reading re-lands, with stable numbers across runs (§5.3).
4. **Upkeep economics.** Rounds per kept interval and the cost of watering, read absolutely between what nothing tends to keep (luck) and what an omniscient tender would have kept (hindsight), with the envelope's failures reported as its edges (§5.4).

What flew is the smallest grammar that exercises the whole loop. Two claim lines and two term lines, one dial, a two-stage path: a proof of concept, deliberately simple. The claims of this paper are about the loop's existence and its honesty — that a promise can be steered, refused, survived, and priced — not about its reach. Scaling is the open direction the boundaries point at: denser maps, deeper corridors, more lines, and eventually several promises sharing one substrate.

One containment rules everything below. The comparison is contained: no external keeper family appears, and the honesty of every number is carried by two internal anchors: what nothing tends to keep (the same promise, nobody tending — is the band satisfied by luck?) and what an omniscient tender would have kept (computed after the fact from the recorded truth). Every kept fraction in this paper reads absolutely between those anchors. A comparison across keeper families is the named follow-up's spine, not a row here; the pricing throughout stays in the promise's own units — rounds and waterings.

2 The world and the grammar

Substrate. The substrate is whatever the agents tend: physical, computational, or mathematical. In this campaign it is a live bench of coupled nodes; each node exposes parameters (dials) and readings (objectives), and a change of one node's dial travels through measured couplings into other nodes' readings. A truth channel records the bench's own state throughout — the answer key. It is never read at decision time; it scores after the fact.

The connectome, and the map. The substrate's coupling structure, once measured edge by edge — each edge with its response curve and error bar, the work of the prior papers — is its *connectome*. From here on, this paper calls the connectome simply *the map*: the thing that is planned through, refused for, and paid for (Figure 1).

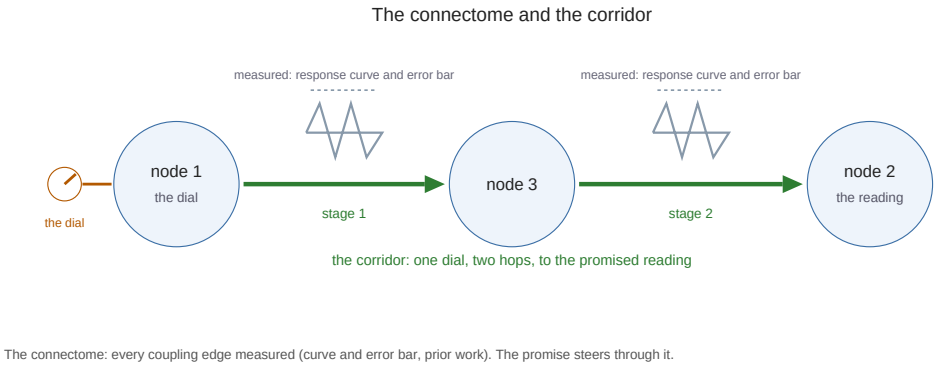


Figure 1: The connectome and the corridor. Nodes of the substrate, coupling edges measured (curve and error bar, prior work). A promise steers through the connectome — here one dial, two stages, to the promised reading.

Agents. The agents are tending agents, never optimizers: they tend a shared substrate. A keeper holds

a promise; scorers witness lines; measurers refresh the map. None of them maximizes anything.

Promise. A promise is a band $[\mathrm{lo}, \mathrm{hi}]$ on a reading, held over time: the delivered reading must sit inside the band for the promise to count as kept in a given interval. Promises conjoin: several lines must hold simultaneously, and the conjunction is exact — one line out ends the flight's landing claim.

The four-line grammar. The campaign's shape is 2+2: two claim lines (bands on two readings, one of them two hops away) and two term lines (bands on related readings that the promise's semantics require to hold). Every line — claim or term — is judged by the same per-line verdict record, with the same bars.

Naming, once. Internally, the held promise of this paper is a *steerwish* — intent made keepable: a promise *to* a live system, declared by its holder and never invented by the engine, kept against drift, and refused — not faked — when the system says no. Its family contains one sibling, the *seekwish* (an exploration wish, which deliberately explores the substrate's range rather than holding a band); it is named here for completeness and claimed nowhere in this paper.

Rounds and waterings. A *round* is one full pass of the keeping loop: plan through the map, act, judge. A *watering* is a discrete corrective intervention, fired when the reading leaves the band by more than continuous compensation can absorb. The loop itself is Figure 2.

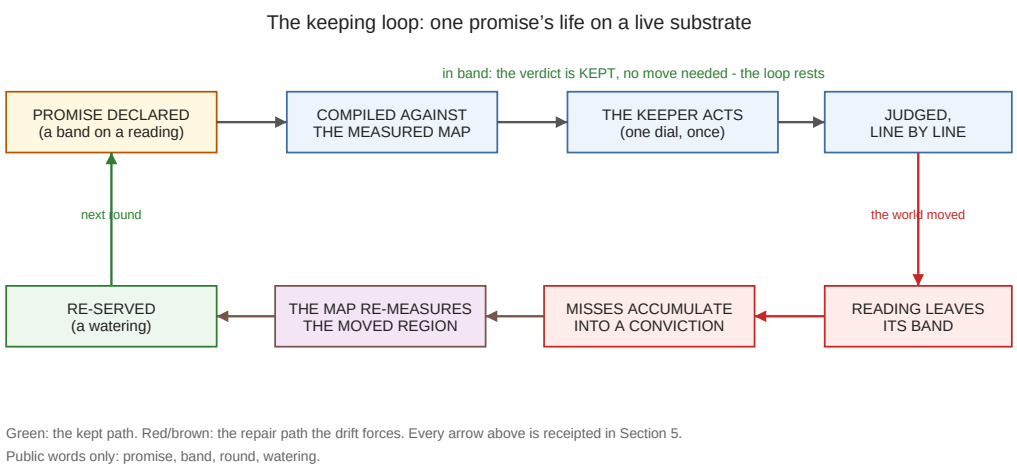


Figure 2: The keeping loop. A promise holds a band on a reading of a shared substrate. The keeper plans through its measured map, acts once, and the per-line verdict records judge the delivered reading against the promised band. Rounds repeat while the promise stands; a watering fires when the world exceeds what compensation absorbs.

The engine under the loop is prior work, used as settled: response curves with per-point bars [2] and their composition across hops [3] — together, the connectome of the preceding paragraph. Neither prior result is re-litigated here; what this paper asks is what the map is worth once a promise must be held on it while the substrate drifts — steering by it, refusing for it, surviving on it, and paying for it.

3 What makes a verdict countable

Every claim in this paper rests on verdicts — landed, missed, refused — so the rules that make a verdict countable are part of the instrument. There are two.

Everything is recorded. Every run writes the same diary — per tick (reading, in band or not, dial position, trial spent) and per event (declared, planned, missed, probed, re-planned, landed, broken, released). The truth channel — the bench's own answer key — records the real state alongside. It is

never read at decision time; it scores after the fact. Verdicts are recomputed from these recordings, never taken from a run's badge or summary.

A verdict needs its witness. Every promise line has its own verdict record, and the record only counts readings taken while a scorer was assigned to that line. The rule is simple: no assignment, no verdict — an unwitnessed line is not a passed line. And a plan that moves dials must carry its receivers, so that every line the promise depends on stays witnessed while the plan executes.

That is the whole section: record everything, and count only what was witnessed.

4 The campaign and the free bracket

The unit of the campaign is a *run*: one prepared session of the live bench — the substrate measured to a finished map (if the cell needs one), the promise declared, the run flown, the run closed. Table 1 gathers every run's shape in one place: the promise, its band, the drift the world was scheduled to deliver, and the window. The scenario details live there and only there; the results sections speak verdicts.

Experiment	The promise	Allowed band	The world's movement	Run length
Steering and its seed repeats	four lines at once: two readings the holder asked about, two supporting readings that must also hold	≈ 0.16	none — the maps were learned before the promise	minutes per flight
Refusal	one target, placed beyond everything the dials were measured to reach	—	none	minutes
Drift battery	one line, held on a finished map	—	one deliberate shape change, mid-run	~35 min each
Upkeep	one line, held for hours	first flight ± 0.15, later flights 0.08	a slow scheduled slide: 0.0008 per step, up to 0.15 in total	6 h, then overnight; the kept hold: four hours

Table 1: How to read this table: each experiment is defined by four things — what was promised, how wide the allowed band was, whether and how the world was made to move, and how long the run lasted. A run is one prepared session of the bench; these four things are everything a re-run needs. A dash means no single number applies. The first upkeep flight's recording ended early on a transport fault (a known class); its verdict reads from the recorded strip to that point.

Every number in this paper reads three ways: if nothing tended, what we tended, and if something omniscient had tended. The first is the floor — the same promise, the same drifting world, nobody acting: whatever stays inside the band is what luck buys. If the floor is high, the band was vacuous and the run certifies nothing; that verdict is itself a receipt. The third is the ceiling — computed after the run from the recorded answer key: what perfect hindsight would have kept. No oracle ever flew live; it is a calculation, not a competitor. Between the two sits what we tended — the keeper's number, readable on its own.

Doctrine, inherited: the sweep is characterization, not pass/fail. Expected degradation is the floor and is reported; unexpected failure is an instrument fault and is cured with receipts (the witnessing repairs of §3 are exactly that class). Boundary rows are receipts, never discards.

5 Results

Each subsection carries its own table: verdicts and margins first, and every caption opens by saying how to read the table. Every run's shape — promise, band, scheduled drift, window — sits in one table in the campaign section (Table 1); the results speak verdicts.

5.1 Steering: the four-line grammar

The four-line conjunction — two *claim* lines (readings the holder asked about) and two *term* lines (supporting readings that must also hold) — flew through one dial and a two-stage path.

The landing receipts. Both term lines judged *and honored live* — 0.8204 $[0.64, 1.00]$ and 0.3534 $[0.16, 0.52]$; one claim line judged missed by 0.022 (predicted 0.6486, delivered 0.5685), stated as its row says. The follow-up hold-shape flight landed with its claims judged (Table 2).

The boundary, as measured. At the campaign's map density the best claim miss is 0.022 against bars near 0.01, and the joint down-shape ($-0.10/-0.06$) was refused by name three times across fresh runs: on coarse maps the composed plan reads conflicts that are not real, and the planner refuses what it cannot compose honestly. This is a boundary of map density — a denser map composes the two-stage path honestly, and the four-of-four landing is one dense map away. Boundary, not failure — and the boundary's direction is known.

Repetition. Five seed runs, four landed (a, c, d, e; Table 3): margins 0.047–0.063 above band floors, two flights including honest undecidable first windows (bars straddling an edge at n=3) before their landing stamps, declare-to-verdict spans of 14–60 s. The fifth (S2-b) missed by 0.004 over the edge — declared four hours after its pre-measurement finished, a timing artifact of the operators, not the engine; the run sat and wandered, and the row says so. The repetition bar (three of five) is cleared (Table 3).

Flight	Outcome	The numbers
The four lines, flown once as the smallest full test	supporting lines landed; one asked-for line missed	supporting lines 0.820 and 0.353, both inside their bands; the missed line: predicted 0.6486, delivered 0.5685 — 0.022 short
The same shape, held again	landed — the asked-for lines judged and held	verdicts recorded in the run's log
A harder ask: two lines moved down together (three fresh runs)	refused, all three times	the map was too coarse: the planner saw conflicts that were not there, and refused rather than guess

Table 2: How to read this table: a line *lands* when its delivered reading sits inside its promised band, and is *judged* when the instrument recorded a verdict for it. The asked-for lines are the paper's claim lines; the supporting lines are its term lines (§2). Each part of the four-line promise is proven here on its own; all four at once is one denser map away (§5.1). Run shape: Table 1, first row.

5.2 Refusal: declining the unmeasurable

A finished-map run; the declared target placed deliberately outside the planner's measured range. The engine refused, by name: *outside the measured range*, with the anatomy named — the hill region $[25, 75]$; the best available branch shift 0.1184 against the required 0.6363; the noise gate 0.0514 (Figure 3). Two candidate paths were evaluated; both refused. Zero actuation: no plan, no assignment, no trial; nothing moved, and the promise row reads unservable (Table 4).

Run	Outcome	How close
a	landed	0.047 and 0.063 above the band's lower edge
b	missed	0.004 over the upper edge — the promise was declared four hours after its preparation measurements (our timing error, not the engine's), and the world had time to wander
c	landed	the first check was too close to call — the measurement's own error bar straddled the edge; the next check sat inside
d	landed	comfortably above the lower edge
e	landed	both lines mid-band

Table 3: How to read this table: the same promise was prepared and flown five times, differing only in random seed; *how close* is the distance between the delivered reading and the nearest band edge. A check is *too close to call* when the measurement's own error bar overlaps that edge. Four land, one misses at the edge. Runs a and e lost their raw timelines at collection — verdicts from the flight log and the record.

Attempt	Why it refused
Final	outside the measured range: the promise demanded the reading move by 0.636; the largest move the measured dials could compose was 0.118; the smallest move the bench can tell apart from noise is 0.051
First	the instrument had no measurement for the line (our setup error) — and a refusal without a measurement proves nothing

Table 4: How to read this table: each row is one attempt to declare a promise the engine judged impossible; *refused* means it named its reason and moved nothing — no plan, no assignment, no trial. The first refusal was the instrument's own fault; the final one is the receipt.

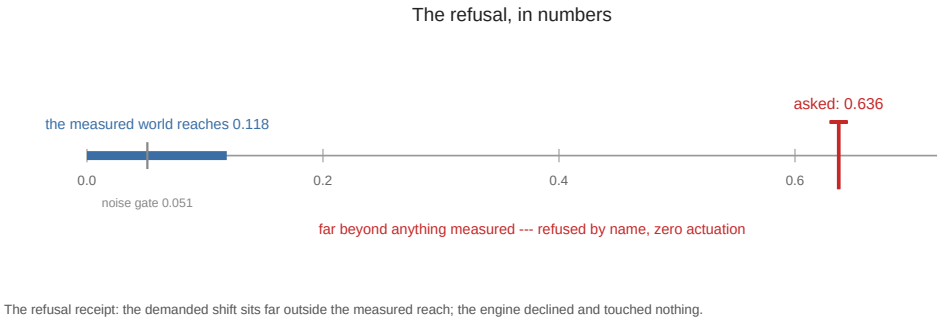


Figure 3: The refusal, in numbers. The demanded shift (0.636) sits far beyond the measured reach (0.118, noise gate at 0.051): the engine declined by name and touched nothing.

The instrument's own trail is part of the receipt. The first attempt in this run also refused — *unmeasured outcome* — and the cause was the operators': a mis-addressed read left the scorer measuring nothing, and a refusal on an unwitnessed line is worth exactly nothing. Banked, fixed, re-flown; both refusals are in the bundle. A refusal semantics is only as good as the witness it refuses on.

This is the epistemic half of the loop: infinity versus zero. The keeper does not extrapolate beyond its measured range — it states what would be required (0.6363), what is available (0.1184), and touches nothing.

5.3 Drift survival: the arc with stable numbers

The battery (five runs, one declared shape change each) repeats one arc with stable numbers: slide to conviction 76–90 s; misses to streak two, plus one undecidable; conviction ratios 65–300× the plan's own error bars; declare to landed 1–2 s.

One run, told fully. The world moves off the finished map. The standing watch fires unprompted — 19 s cue-to-alarm, reading 0.0618 past a 0.0500 band, a second sample at 0.1557 — and the targeted probe returns the first usable hint in the instrument's history, 76 s after the slide cue. The measurement re-opens around the moved region; the refresh is physical (the probing cycle is visible on the strip); the reading re-lands; unattended regions heal. The conviction itself is priced: predicted shifts with their bars against delivered readings land 65–300× the plan's own bars — far enough out that drift, not noise, is the verdict. Figure 4 shows the arc in two views on one timeline: above, the promise's experience — the reading against its band, quiet, then the break, then re-landed; below, the map's experience — its error against the answer key, collapsing at the re-measure. The stable numbers sit in Table 5.

What was measured	What the five runs showed
From the world's move to the stale-map verdict	76–90 s
Misses before that verdict	two (one too close to call)
How far the missed readings landed	65–300× the plan's own error bar
From promise declared to promise landed	1–2 s
The whole cycle happening a second time	once — a single-run receipt

Table 5: How to read this table: in each of the five runs the world was moved mid-run while a promise stood; these are the timings of what followed, stable across all five. A *miss* is a judged check that found the reading outside its band; the *stale-map verdict* is the engine's own judgment that its learned map no longer matched the world; the *plan's own error bar* is the uncertainty the map claimed for itself at planning time.

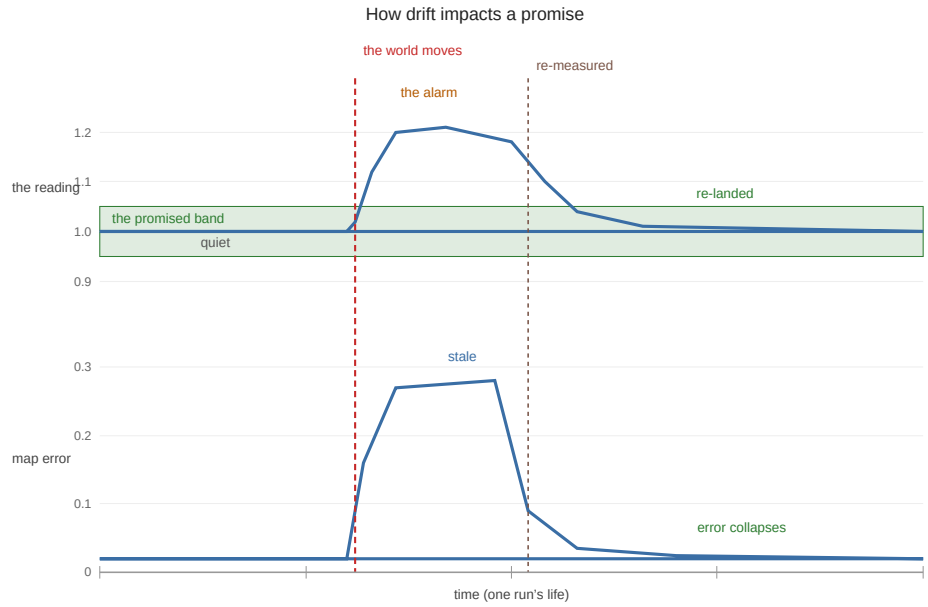
The re-learn limit, measured. At walking pace, a full 101-level grid refresh covers ≈20 levels in 13 minutes — a full re-learn outlives the run's window. Re-learning starves behind successful-looking re-plans (109 misses against 116 re-plans in the starved era) until the miss streak breaks the treadmill and the refresh fires. The honest consequence: the full second lap (re-learn, re-serve, re-miss) has a single-run receipt (n=1), and is framed done on that receipt — one sentence, no further flights.

5.4 Upkeep: what keeping costs

The message of this section: holding a promise on a moving world costs work, and the cost can be measured. One number carries it. Over four hours of a slowly sliding world, the keeper kept the reading inside its band for 0.874 of the judged windows — doing nothing would have kept 0.013, perfect foresight 1.000 (Table 6, Figure 6). Figure 5 shows the recorded hold itself: every judged window of the four hours.

The price behind the number is continuous work: the keeper served trials for the whole four hours, compensating as the world slid. It never lost the band outright — zero misses; the windows it did not keep were minutes spent riding the band's edge while the world wandered. The promise's own claim line judged landed throughout, median reading 0.98. A *watering* — a discrete corrective move — never became necessary: compensation never fell that far behind.

Two earlier flights taught how to read the number. In the first, the band was too wide (±0.15 against a 0.15 slide): doing nothing also kept everything (1.000), so the keeper's 0.993 meant nothing — a satisfied luck floor certifies nothing. In the second, the band was narrow but placed on a fleeting reading (an operator error, kept on the record): the world outran the compensation — sixteen breaks,



Schematic of the documented battery arc (Section 5.3).

Figure 4: How drift impacts a promise — two views of one arc (schematic, from the documented battery arc; numbers in Table 5). Above: the reading against its promised band — quiet, the world moves, the break is the alarm (far beyond the plan's own error bar, so drift not noise), the moved region re-measured, re-landed. Below: the same arc as the map's error against the answer key — the maintained property, collapsing at the re-measure.

kept only 0.093. Those two failures are the edges of the economics; the third flight is the keep between them.

Why is perfect foresight exactly 1.000? With the slide known in advance, every window can be compensated before it happens — foresight is never surprised. The keeper's gap to it (0.126) is the measured price of acting on a map instead of knowing the future.

Flight	Kept	Luck alone	Foresight
First — wide band (±0.15)	0.993	1.000	not scored
Second — narrow band, wrong place	0.093	0.000	not scored
Third — narrow band, right place	**0.874**	0.013	1.000

Table 6: How to read this table: *kept* is the share of judged windows — the run is scored at regular intervals, not continuously — that the reading spent inside its band; *luck alone* is that share with nobody acting; *foresight* is calculated after the run from the recorded answer key — what perfect foreknowledge would have kept — and never flew. Foresight was not scored for the two teaching flights. One slide, three band choices: a wide band makes keeping meaningless (luck alone kept everything); a narrow band in the wrong place makes it impossible; the third is the honest keep.

The economics, stated in the promise's own units. The keep's price is the continuous hold: trials served for four hours at the run's per-trial cost, buying 0.87-kept against a 0.01 floor. Discrete waterings fire when the world exceeds what continuous compensation absorbs — v2's sixteen breaks are that boundary's other side. (Figure 6.)

Why hindsight is exactly 1.000*, and what the gap means.* With the drift path known in advance, every window can be pre-compensated; perfect foresight is never surprised, so it keeps all windows — the ceiling is a structural 1.000, not an achievement. The keeper's gap to it (0.126) is the measured price

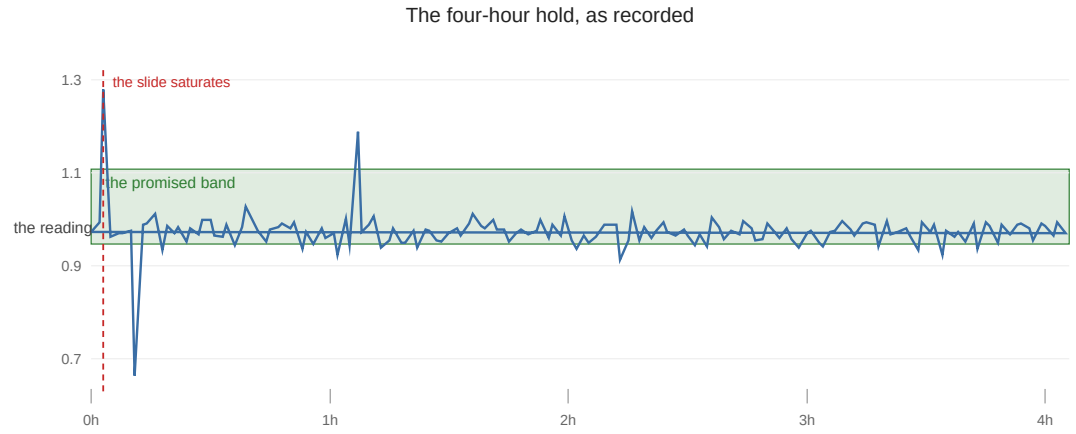


Figure 5: The four-hour hold, as recorded: the reading at every judged window (159 samples from the strip log) against its promised band. Kept 139 of 159; the excursions are the edge-riding straddles; the slide saturates minutes in.

of acting on a map instead of a crystal ball: the keeper never broke — zero misses — but it rides the edge while the world wanders, and edge-riding costs windows. For contrast, v2's gap was 0.907: there the world outpaced compensation, and that is what a bad gap looks like. The oracle is a ceiling by construction, not an arm that flew; it bounds what any keeper could have done with the truth in hand.

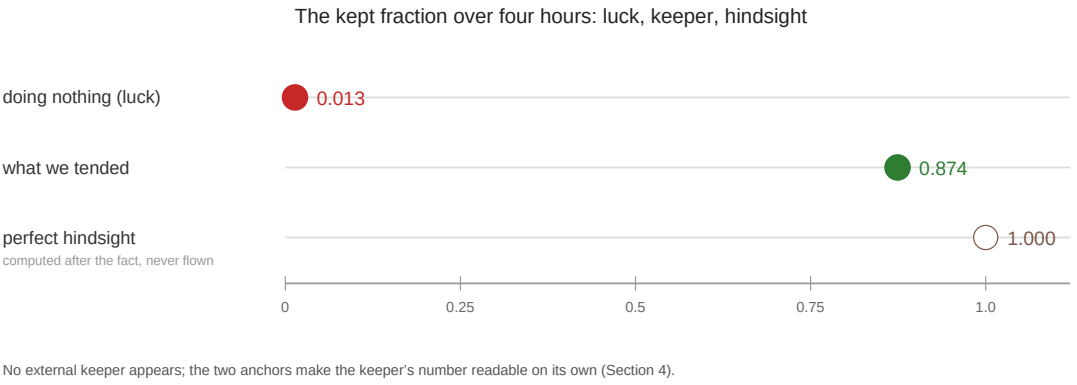


Figure 6: The kept fraction over four hours, on one scale: what luck keeps (0.013), what we tended (0.874), what perfect hindsight computes after the fact (1.000, never flown).

6 Discussion

What the map is worth under drift. The worth of a maintained map is read absolutely: between a floor that says what luck buys (0.013) and a ceiling that says what omniscience buys (1.000), the keeper holds 0.874 for four hours through a saturating slide. The renewal receipt certifies the *maintained* property live against ground truth — map error jumps at the break and collapses at the refresh — and it costs nothing extra: it is a byproduct of runs the campaign already flew.

Why not just optimize, one paragraph. A maximizer and a keeper share machinery — both plan through a model — but their objects differ. Pointed at a substrate, a maximizer drives to an optimum and structurally cannot comply with a promise whose band sits away from it: a promise is a chosen state, not an optimum. The nearest ancestor of this paper's refusal is safe model-based optimization with

Gaussian processes — the safe-optimization line bounds exploration to a trusted region — but its stance is more permissive than the one receipted here: those keepers may act inside the trusted set, while this keeper declined the target itself, by name, with zero actuation. Adaptive model-predictive control re-learns moving plants toward references; it does not price upkeep in the plant's own units or refuse the unmeasurable as a first-class verdict.

The boundaries, collected. The four-line landing is gated on map density (best claim miss 0.022 vs bars ≈ 0.01; three named refusals of the joint down-shape); the upkeep economics have two measured edges (a vacuous band, a wrong cut) and one kept middle; full grid re-learn outlives the run window at walking pace; the second lap rests on a single receipt. And what flew is the smallest grammar that exercises the whole loop — scaling (denser maps, deeper corridors, more lines, simultaneous promises) is the named direction. A journal-only literature residual sits outside arXiv's reach — noted in the writing-day sweep, not blocking.

What this paper deliberately does not contain. No external keeper, no comparative language, no scoreboard. The honesty is carried by the free bracket and by rows that report their own failures. And the pricing stays in the promise's own units — rounds and waterings — by design.

Future work

Scaling. What flew is the smallest grammar that exercises the whole loop. The open directions are the ones the boundaries point at: denser maps, deeper corridors, more lines, and eventually several promises sharing one substrate.

The comparison. This paper reads every number between internal anchors — luck and foresight. The natural next question is how the same loop fares against external keeper families on the same runs. That comparison is the follow-up paper's question, and it stands on these receipts.

7 Related work

Related work is context, not a wall; the phrasing below is deliberately defensive. Each line holds a piece of what this paper assembles — no single one holds the assembly.

Careful optimizers. A family of methods explores a system cautiously: they only try settings their internal model already considers safe, and expand slowly. Representative work optimizes quadrotor controllers through Gaussian-process models [4], proves probably-safe variants of that idea [5], brings trust-region safety to automotive control [6], and adds safety to a classic evolutionary optimizer [7]. All of them bound *exploration* — where they may poke next. The refusal receipted here is the stricter stance: the target itself is declined, by name, with zero actuation, and the one declining owns actuators.

Controllers that re-learn. Adaptive model-predictive control [8] and its modern stochastic variants [9] keep their model of a changing plant up to date and steer toward a setpoint. They lack the promise grammar — bands as first-class objects, conjunctions, a named refusal of the unmeasurable — and they do not price the upkeep.

The economics of promising. Work on commitment devices in cooperative AI [10] studies when agents should bind themselves to promises, in terms of norms and incentives. This paper prices the keeping — against a substrate, with measurements.

The cost of keeping systems alive. Self-adaptive software [11, 12] and the recent mapping of agentic technical debt — the accumulating cost of keeping AI-based systems viable [13] — name these costs. None measure them against a luck floor and a hindsight ceiling.

Systems that decline to answer. Machine learning has a reject option: classifiers and regressors that say "not me" when unsure [14, 15, 16], and safe exploration that declines unsafe states [17]. All decline

answers. Here the one declining owns actuators and stops the machinery — an action declined, with the reason named and the anatomy priced.

To our knowledge, no neighbor does what this loop does together: keep a declared promise on a live, drifting substrate, refuse by name what it cannot measure, and price the keeping between luck and perfect hindsight. We are not aware of anyone else this close.

8 Conclusion

Four kept promises. The map steers: the same scenario landed again and again, and the one boundary is a map-density limit with a known cure. The map refuses: asked for a target beyond its measured range, it declined by name and touched nothing. The map survives drift: the world moved, the misses were counted, the map was refreshed, the reading returned. And the keeping is priced: between what luck keeps and what foresight would have kept, the keeper held most of it, for four hours, without a single break. A held promise on a drifting substrate is a measurable, refusable, payable thing. What flew is the smallest grammar that exercises the whole loop; scaling is the named direction, and the comparison across keeper families is the follow-up's question — both stand on these receipts.

Reproducibility

The campaign flew on a self-hosted live bench — the estate's generic mathematical bench, the same bench family the prior papers measured. Its logic is simple: nodes expose parameters (dials) and return readings; planted couplings — chosen shapes, such as saturating or tent maps, at chosen strengths — carry one node's dial changes into another node's reading; a drift schedule moves the planted values over the run's life; and a closed-form truth channel records what was actually planted at every tick. Nothing planted is visible to the keeper: it learns the couplings by probing, exactly as in the prior papers; the truth channel exists only to score. The run shapes of Table 1 — promise, band, scheduled drift, window — are then the full scenario definitions.

Each run's record is retained in the evidence bundles — the per-tick diary, the per-line verdict cards, the reading strips, the truth and drift recordings — except where cards did not survive collection; those runs are carried by their documented numbers, marked as such in the figures and tables. Every figure regenerates from its bundle by the committed scripts, and the upkeep bracket — floor, kept fraction, oracle ceiling — recomputes from the strip log alone; verdicts are recomputed from curves and strips, never read from run badges. Full independent reproduction requires this estate's stack; on a compatible bench, Table 1 plus the bundles are everything a re-run needs.

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